Multimedia Signal Processing - 2nd Module

July, 26 2021

1. **Question (8 pts)** Consider a continuous source X that emits the following sequence of symbols

$$x = \{1.5,\ 3,\ 4,\ 7,\ 2,\ 3.5,\ 6,\ 0\}.$$

Consider a scalar quantizer with the input-output relation expressed by

$$y = q(x) = \Delta \left\lceil \frac{x}{\Delta} \right\rceil,$$

where $\Delta = 2$ and $\lceil \cdot \rceil$ is the "ceiling" operator that maps x to the least integer greater than or equal to x.

   (a) Quantize the sequence x to obtain y.

   (b) Compute the mean square error (MSE) introduced by the quantizer.

   (c) Estimate the pmf of the source Y modeling the quantizer output considering the sequence y as a representative example.

   (d) Compute the entropy $H(Y)$ of the source Y.

   (e) Derive the Huffman code for the sequence y, showing a picture of the Huffman tree.

   (f) Compute the average code length of the Huffman code.

   (g) Encode the sequence y with the derived Huffman code (i.e., write the extension C^*).

2. **IF ONLINE: Question (7 pts)**
Oral examination.

3. **IF IN CLASS: Question 4 pts)** Consider the problem of lossy coding through quantization.

   - Define a scalar quantizer.
   - Explain how to apply Lloyd-Max quantization.

4. **IF IN CLASS: Question (3 pts)** Consider a color image in the RGB color space. Clearly explain how to encode and decode the image using the JPEG lossy coding standard.

5. **Questions (8 pts - each answer can be either TRUE or FALSE - Incorrect answers give you -1 points, unanswered 0 points, correct answers 1 point - Provide a brief comment when in doubt)**

   - [T] [F] Given two independent discrete memoryless sources (DMS) X and Y, it is possible to infer the joint entropy $H(X,Y)$ from the knowledge of the marginal entropy $H(X)$ and $H(Y)$.
   - [T] [F] It is possible to find a one-to-one code that is not an instantaneous code.
   - [T] [F] Coding gain analysis shows that using a DPCM may provide more distortion than PCM at the same rate in some situations.
   - [T] [F] Consider transform coding. Any square matrix can be used as transform.
   - [T] [F] A DMS source with an alphabet composed by two symbols can have entropy equal to 1.
   - [T] [F] The average code length provided by Huffman coding equals that of a Shannon code for dyadic sources (i.e., sources with p.m.f. described by negative powers of 2).
   - [T] [F] In speech coding, the coefficients of the filter estimated using LPC are typically directly quantized.
   - [T] [F] Given a DMS source X, it is not possible to encode symbols of X with a rate greater than $H(X)+1$.

6. **Question (Matlab - 10pts)**

   Load the default Matlab image `peppers.png` using the command `imread('peppers.png')`.

   (a) Display the image.

   (b) Extract the green color plane.

   (c) Considering the green image component, split it into 8×8 non-overlapping blocks.

   (d) Apply 2D DCT to each block.

   (e) Compute and display the histogram of the DCT coefficients in position $(1,1)$ (i.e., the DC) across the blocks.

   (f) Set the DCT coefficient in position $(1,1)$ equal to zero for all blocks.

   (g) Apply inverse 2D DCT to the modified blocks.

   (h) Put the blocks together to recreate an image the same size of the original green component.

   (i) How do you expect this image should look like compared to the original one?